Pro Football Sports Science Research Update

October 2023

Characterizing neck injuries in the National Football League: A descriptive epidemiology study. BMC Musculoskeletal Disorders, 24:702, 2023.

Chronic changes in muscle architecture and aponeurosis structure following calf muscle strain injuries. Scandinavian Journal of Medicine and Science in Sports, Ahead of Print: 1-13, 2023.

Considerations for the consumption of vitamin and mineral supplements in athlete populations. Sports Medicine, Ahead of Print: 1-10, 2023.

Estimating the relationship between the symptom-free waiting period and injury rates after return to play from concussion: A simulation analysis using CARE Consortium data. Sports Medicine, Ahead of Print: 1-16, 2023.

Evidence-based nutritional approaches to enhance exercise adaptations. Current Opinion in Clinical Nutrition, Ahead of Print: 1-7, 2023.

Flight path and scheduling effects on perceived jet lag, fatigue, and sleep in footballers traveling to and from National teams. International Journal of Sports Physiology and Performance, Ahead of Print: 1-9, 2023.

Hamstring strain injury risk factors in Australian football change over the course of the season. Medicine and Science in Sports and Exercise, Ahead of Print: 1-45, 2023.

Imaging-detected sports injuries and imaging-guided interventions in athletes during the 2022 FIFA soccer World Cup. Skeletal Radiology, Ahead of Print: 1-10, 2023.

"Is it a slow day or a go day?": The perceptions and applications of velocity-based training within elite strength and conditioning. International Journal of Sports Science & Coaching, 18(4): 1217-1228, 2023.

Media's effect on athletes' mental health. Clinics in Sports Medicine, Ahead of Print: 1-12, 2023.

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Metacarpal fractures in the National Football League: Injury characteristics, management, and return to play. Journal of Hand Surgery, 48(6):626.e1-626.e8, 2023.

Monitoring external workload with wearable technology after anterior cruciate ligament reconstruction – A scoping review. The Orthopaedic Journal of Sports Medicine, 11(8): 1-10, 2023.

Performance-enhancing drugs in healthy athletes: An umbrella review of systematic reviews and meta-analyses. Sports Health, Ahead of Print: 1-11, 2023.

Plasma myoglobin indicates muscle damage associated with acceleration/deceleration during football. The Journal of Sports Medicine and Physical Fitness, Ahead of Print: 1-6, 2023.

Postinjury alcohol use is associated with prolonged recovery after concussion in NCAA athletes. Clinical Journal of Sport Medicine, Ahead of Print: 1-8, 2023.

Prophylactic knee bracing in offensive linemen of the National Football League – A retrospective analysis of usage trends, player performance, and major knee injury. The Orthopaedic Journal of Sports Medicine, 11(8): 1-8, 2023.

Proximal rectus femoris tendon tear in a professional football placekicker: A case report. Current Sports Medicine Reports, Vol. 22(6): 187-190, 2023.

Return to play following injuries in pro football (soccer): Insights into the real-life practices of 85 elite practitioners around diagnostics, progression strategies and reintegration processes. Sport Performance Science Reports, Vol. 1: 1-20, 2023.

Return-to-play for elite athletes with genetic heart diseases predisposing to sudden cardiac death. Journal of the American College of Cardiology, Vol. 82(8): 661-670, 2023.

SARS-CoV-2 infection increases the risk of muscle injury in professional male soccer players – A retrospective analysis of the Italian and Spanish major leagues. Infectious Disease Reports, 15: 425-435, 2023.

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Vitamin C supplementation and athletic performance: A review. Current Sports Medicine Reports, Vol. 22(7): 255-259, 2023.

The impact of daytime napping following normal night-time sleep on physical performance: A systematic review, meta-analysis and meta-regression. Sports Medicine, Ahead of Print: 1-23, 2023.

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Characterizing neck injuries in the National Football League: A descriptive epidemiology study. BMC Musculoskeletal Disorders, 24:702, 2023.

Background: Neck injury is a common and often debilitating injury among athletes participating in American football. Limited data exists regarding neck injuries among elite athletes in the National Football League (NFL). To characterize the epidemiology of non-season ending, season-ending, and career-ending neck injuries in the NFL from 2016 through 2021.

Methods: Athletes who sustained neck injuries were identified using the NFL's injured reserve (IR) list between the 2016 and 2021 seasons. Demographics and return to sport (RTS) data were collected. Available game footages were reviewed to identify the mechanism of injury (MOI). Injury incidence rates were calculated based on per team play basis.

Results: During the 6-year study period, 464 players (mean age 26.8±3.2 years) were placed on the injury reserve list due to neck injuries. There were 285 defensive players and 179 offensive players injured (61.4 vs 38.6%, respectively, $p<0.001$). Defensive back was the most common position to sustain a neck injury (111 players, 23.9%). 407 players (87.7%) sustained non-season-ending injuries with a mean RTS at 9.2±11.3 days. 36 players (7.8%) sustained season-ending injuries with a mean RTS at 378.6±162.0 days. 21 players (4.5%) sustained career-ending injuries. The overall incidence of neck injuries was 23.5 per 10,000 team plays. The incidence of season-ending injuries and career-ending injuries were 1.82 and 1.06 per 10,000 team plays, respectively. There were 38 injuries with available footages for MOI assessment (23 non-season-ending, 9 season-ending, 6 career-ending). Head-to-head contact was seen in 15 injuries (39.5%), head-down tackling in 11 injuries (28.9%), direct extremity-to-head contact in 7 injuries (18.4%), and head-to-ground contact in 5 injuries (13.2%). There was no significant difference in age, position, or MOI among players sustaining non-season-ending, season-ending, and career-ending injuries.

Conclusion: There is a high incidence of neck injuries among NFL athletes with predictable MOIs including head-to-head contact, head-down tackling, direct extremity-to-head contact, and head-to-ground contact. Defensive players were more likely to sustain neck injuries compared to offensive players. Defensive back was the most common position to sustain a neck injury.

Level of evidence: III.

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Chronic changes in muscle architecture and aponeurosis structure following calf muscle strain injuries. Scandinavian Journal of Medicine and Science in Sports, Ahead of Print: 1-13, 2023.

Background: Muscle strain injuries in the human calf muscles are frequent sports injuries with high recurrence. Potential structural and functional changes in the medial head of the musculus gastrocnemius (GM) and the associated aponeurosis are not well documented.

Purpose: To test whether a GM muscle strain injury affects muscle fascicle length, pennation angle, and the morphology of the deep aponeurosis at rest and during muscle contraction long time after the injury. Additionally, electromyography (EMG) of the GM and the soleus muscle during a unilateral heel rise was measured in the injured and uninjured calf.

Methods: GM fascicle length, pennation angle, and aponeurosis thickness was analyzed on dynamic ultrasonography (US) recordings in 10 participants with a chronic calf strain. In addition, US images taken across the distal portion and mid-belly of the GM were analyzed at three different ankle positions. EMG recordings were obtained during a unilateral heel rise.

Results: The pennation angle of the injured distal GM was significantly larger compared to the uninjured GM in the contracted, but not the relaxed state. Pennation angle increased more in the injured compared to the uninjured GM during contraction. Fascicle length was shorter in the most distal portion of the injured GM. Fascicles at the distal portion of the injured GM showed a pronounced curvilinear shape as the muscle contracted and the aponeurosis was enlarged in the injured compared to the uninjured GM. The ratio between GM and soleus EMG activity showed a significantly higher relative soleus activity in the injured compared to the healthy calf.

Conclusion: The greater change in pennation angle and curvilinear fascicle shape during contraction suggest that a long-term consequence after a muscle strain injury is that some muscle fibers at the distal GM are not actively engaged. The significantly enlarged aponeurosis indicates a substantial and long-lasting connective tissue involvement following strain injuries.

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Considerations for the consumption of vitamin and mineral supplements in athlete populations. Sports Medicine, Ahead of Print: 1-10, 2023.

Abstract - Vitamins and minerals are of fundamental importance to numerous human functions that are essential to optimize athlete performance. Athletes incur a high turnover of key vitamins and minerals and are therefore dependent on sufficient energy intake to replenish nutrient stores. However, many athletes are poor at servicing their energy replenishment needs, especially female athletes, and although a 'food first approach' to meeting nutrient requirements is the primary goal, it may be important for some athletes to consider a vitamin and/or mineral supplement to meet their daily needs. When working to determine if an athlete requires vitamin or mineral supplements, practitioners should use a robust framework to assess the overall energy requirements, current dietary practices and the biological and clinical status of their athletes. Of note, any supplementation plan should account for the various factors that may impact the efficacy of the approach (e.g. athlete sex, the nutrient recommended dietary intake, supplement dose/timing, co-consumption of other foods and any food–drug interactions). Importantly, there are numerous vitamins and minerals of key importance to athletes, each having specific relevance to certain situations (e.g. iron and B vitamins are significant contributors to hematological adaptation, calcium and vitamin D are important to bone health and folate is important in the female athlete); therefore, the appropriate supplement for a given situation should be carefully considered and consumed with the goal to augment an athlete's diet.

Key Points

- Athletes are generally poor at servicing their energy intake requirements, and although a 'food first approach' is a primary goal, certain situations (i.e. food-restricted diets, poor recovery timing and nutrition, etc.) may warrant an athlete to consider a vitamin or mineral supplement to meet their daily needs.
- There are numerous vitamins and minerals of key importance to athletes, each having specific relevance to certain situations. The usefulness of supplementation for a given situation should be carefully considered before use and consumed to augment an athlete's typical diet.
- Female athletes have nuanced vitamin and mineral requirements that differ to their male counterparts, and the research landscape requires significant work to better understand the unique nutritional challenges they face.

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Estimating the relationship between the symptom-free waiting period and injury rates after return to play from concussion: A simulation analysis using CARE Consortium data. Sports Medicine, Ahead of Print: 1-16, 2023.

Background: A key component of return-to-play (RTP) from sport-related concussion is the symptom-free waiting period (SFWP), i.e., the period during which athletes must remain symptom-free before permitting RTP. Yet, the exact relationship between SFWP and post-RTP injury rates is unclear.

Objective: We design computational simulations to estimate the relationship between the SFWP and rates of repeat concussion and non-concussion time-loss injury up to 30 days post-RTP for male and female collegiate athletes across 13 sports.

Methods: We leverage N=735 female and N=1,094 male post-injury trajectories from the National Collegiate Athletic Association-Department of Defense Concussion Assessment, Research, and Education Consortium.

Results: With a 6-day SFWP, the mean [95% CI] rate of repeat concussion per 1,000 simulations was greatest in ice hockey for females (20.31, [20.16, 20.46]) and American football for males (24.16, [24.05, 24.28]). Non-concussion time-loss injury rates were greatest in field hockey for females (153.66, [152.59, 154.74]) and wrestling for males (247.34, [246.20, 248.48]). Increasing to a 13-day SFWP, ice hockey for females (18.88, [18.79, 18.98]) and American football for males (23.16, [23.09, 24.22]) exhibit the greatest decrease in repeat concussion rates across all sports within their respective sexes. Field hockey for females (143.24, [142.53, 143.94]) and wrestling for males (237.73, [236.67, 237.90]) exhibit the greatest decrease in non-concussion time-loss injury rates. Males receive marginally smaller reductions in injury rates for increased SFWP compared to females (OR = 1.003, $p \leq 0.002$).

Conclusion: Longer SFWPs lead to greater reductions in post-RTP injury rates for athletes in higher risk sports. Moreover, SFWPs should be tailored to sport-specific post-RTP injury risks.

Key Points

- The symptom-free waiting period should be tailored to each individual athlete's risk for injury after return-to-play.
- Athletes with a higher risk of injury after return-to-play achieve greater benefit from longer symptom-free waiting periods compared with athletes with a lower risk of injury.

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- Our study findings can be used to help clinicians select a symptom-free waiting period based on their risk tolerance and athlete-specific injury risks.

Evidence-based nutritional approaches to enhance exercise adaptations. Current Opinion in Clinical Nutrition, Ahead of Print: 1-7, 2023.

Purpose of review: The purpose of this opinion paper is to provide current-day and evidence-based information regarding dietary supplements that support resistance training adaptations or acutely enhance strength-power or endurance performance.

Recent findings: Several independent lines of evidence support that higher protein diets, which can be readily achieved through animal-based protein supplements, optimize muscle mass during periods of resistance training, and this likely facilitates strength increases. Creatine monohydrate supplementation and peri-exercise caffeine consumption also enhance strength and power through distinct mechanisms. Supplements that favorably affect aspects of endurance performance include peri-exercise caffeine, nitrate-containing supplements (e.g., beet root juice), and sodium bicarbonate consumption. Further, beta-alanine supplementation can enhance high-intensity endurance exercise efforts.

Summary: Select dietary supplements can enhance strength and endurance outcomes, and take-home recommendations will be provided for athletes and practitioners aiming to adopt these strategies.

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Flight path and scheduling effects on perceived jet lag, fatigue, and sleep in footballers traveling to and from National teams. International Journal of Sports Physiology and Performance, Ahead of Print: 1-9, 2023.

Purpose: This study examined post-travel perceptual responses of national-team footballers (soccer) following different flight paths, arrival/departure times, and trip contexts.

Methods: Details of 396 flights from national-team players (N = 68) were obtained and verified via an online flight database. Each player provided ratings of perceptual fatigue, sleep, soreness, stress, and jet lag for 2 days before and after each trip. The flight path (continents of departure and arrival), travel context (into vs out of national team), and arrival and departure times were obtained for each trip. Linear mixed models compared the pretravel with post-travel change in perceptual responses based on flight path, context, and schedule.

Results: Perceived jet-lag ratings were more responsive to travel variables ($R2 = .48$) than other perceptual ratings ($R2 < .26$). Travel from Asia to Europe ($P < .05$) and Europe to Australia ($P < .001$) had significantly higher jet-lag ratings than all other paths. Fatigue scores were worst following Asia to Europe ($P < .05$) and Europe to Australia ($P < .05$) travel, while sleep scores were worst following Europe to Australia travel ($P < .01$). Perceptual responses were poorer following travel from national team to club compared with all other travel contexts ($P < .05$). Arrival during the daytime (11 AM to 5 PM) resulted in better perceptual responses than early-morning or late-night arrivals ($P < .05$).

Conclusions: Perceived jet-lag ratings are more responsive to travel demands than perceptual wellness scales in national-team football athletes. Poorer perceptual responses may be expected when travel is longer in nature, ends later in the day, or involves travel out of the national team back to club.

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Hamstring strain injury risk factors in Australian football change over the course of the season. Medicine and Science in Sports and Exercise, Ahead of Print: 1-45, 2023.

Background/aim: To determine which factors were most predictive of hamstring strain injury (HSI) during different stages of the competition in professional Australian Football.

Methods: Across two competitive seasons, eccentric knee flexor strength and biceps femoris long head (BFlh) architecture of 311 Australian Football players (455 player seasons) were assessed at the start and end of pre-season and in the middle of the competitive season. Details of any prospective HSIs were collated by medical staff of participating teams. Multiple logistic regression models were built to identify important risk factors for HSI at the different time points across the season.

Results: There were 16, 33 and 21 new HSIs reported in preseason, early in-season, and late in-season, respectively across two competitive seasons. Multivariate logistic regression and recursive feature selection revealed that risk factors were different for pre-season, early in-season and late in-season HSIs. A combination of prior HSI, age, height and muscle thickness were most associated with pre-season injuries (median AUC, 0.83). Pennation angle and fascicle length had the strongest association with early in-season injuries (median AUC, 0.86). None of the input variables were associated with late in-season injuries (median AUC, 0.46). Identification of early in-season HSIs and late in-season HSIs was not improved by the magnitude of change of data across pre-season (median AUC, 0.67).

Conclusions: Risk factors associated with prospective HSIs were different across the season in Australian Rules Football, with non-modifiable factors (prior HSI, age, and height) mostly associated with pre-season injuries. Early in-season HSIs were associated with modifiable factors, notably BFlh architectural measures. The prediction of in-season HSIs was not improved by assessing the magnitude of change in data across pre-season.

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Imaging-detected sports injuries and imaging-guided interventions in athletes during the 2022 FIFA soccer World Cup. Skeletal Radiology, Ahead of Print: 1-10, 2023.

Objective: To describe imaging-detected musculoskeletal injuries and image-guided interventional procedures during the 2022 FIFA football (soccer) World Cup.

Materials and methods: Retrospective analysis of all radiologic examinations performed in a central medical facility for athletes was performed by two board certified musculoskeletal radiologists. Data on muscle, tendon, ligament, cartilage, and bone injuries were collected according to imaging modality and body part.

Results: A total of 143 radiology examinations in 94 athletes were evaluated at the central medical facility. Magnetic resonance imaging (MRI) was the most utilized modality (67%), followed by radiography (12%), ultrasonography (9%), and computed tomography (4%). Image-guided interventions corresponded to 8% of all radiological examinations. There were 112 injuries described, affecting muscles and tendons (42%), ligaments (25%), cartilage (21%), and bone (12%). Most injured body parts were thigh (27%), foot and ankle (23%), knee (23%), and hip/groin (8%). Most injured players were within the age range of 24–35 years old (71%).

Conclusion: Imaging was utilized in 11% of players who participated in the 2022 FIFA World Cup in Qatar. MRI was the most utilized modality, and acute muscle tears were the most diagnosed type of injury.

Summary: Diagnostic imaging played an important role in diagnosing sports-related injuries during the 2022 FIFA World Cup.

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"Is it a slow day or a go day?": The perceptions and applications of velocity-based training within elite strength and conditioning. International Journal of Sports Science & Coaching, 18(4): 1217-1228, 2023.

Abstract - Velocity-based training (VBT) is a contemporary prescriptive, programming, and testing tool commonly utilized in strength and conditioning (S&C). Over recent years, there has been an influx of peer-reviewed literature investigating several different applications (e.g. load-velocity profiling, velocity loss, load manipulation, and reliability of technology) of VBT. The procedures implemented in research, however, do not always reflect the practices within applied environments. The aim of this study, therefore, was to investigate the perceptions and applications of VBT within elite S&C to enhance contextual understanding and develop appropriate avenues of practitioner-focused research. Fourteen high-performance S&C coaches participated in semi-structured interviews to discuss their experiences of implementing VBT into their practices. Reflexive thematic analysis was adopted, following an inductive and realist approach. Three central organizing themes emerged: Technology, applications, and reflections. Within these central themes, higher order themes consisting of drivers for buying technology; programming, testing, monitoring, and feedback; and benefits, drawbacks, and future uses also emerged. Practitioners reported varied drivers and applications of VBT, often being dictated by simplicity, environmental context, and personal preferences. Coaches perceived VBT to be a beneficial tool yet were cognizant of the drawbacks and challenges in certain settings. VBT is a flexible tool that can support and aid several aspects of S&C planning and delivery, with coaches valuing the impact it can have on training environments, objective prescriptions, tracking player readiness, and program success.

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Media's effect on athletes' mental health. Clinics in Sports Medicine, Ahead of Print: 1-12, 2023.

KEY POINTS

- Athletes potentially face punitive measures for not complying with their sports media obligations.
- Compared to the general population, athletes have additional incentives to benefit from social media use.
- Athletes of diverse and minoritized backgrounds may be subject to marginalizing media coverage.
- Although media attention for athletes has various positive impacts, it can also serve as a profound source of stress and lead to negative mental health consequences.
- Understanding the potential negative impact of media on athletes' mental health may lead sports medicine medical staff to create ways to support the athlete's interactions with media through policies and procedures.

Metacarpal fractures in the National Football League: Injury characteristics, management, and return to play. Journal of Hand Surgery, 48(6):626.e1-626.e8, 2023.

Purpose: This study investigated metacarpal fracture occurrences, characteristics, treatments, and return-to-play times for National Football League (NFL) athletes.

Methods: NFL players who sustained metacarpal fractures during the 2012 to 2018 seasons were reviewed. All players on the 32 NFL team active rosters with metacarpal fractures recorded through the NFL Injury Database were included. Player age, time in the league, player position, injury setting, injury mechanism, fractured ray, management, and return-to-play were recorded.

Results: There were 208 injury occurrences resulting in 1 or more metacarpal fractures, identified in 205 players. Of these, 81 (39%) injuries were operated. Return-to-play data were available for 173 (83%) injured players. The median return-to-play time for all athletes was 15 days (interquartile range, 1-55 days). Of the injured players, 130 (71%) missed time but returned the same season. Within this 130-

player subset, 69 (53%) were treated non-surgically and 61 (47%) operatively with median return-to-play times of 16 days (interquartile range, 6-30 days) and 20 days (interquartile range, 16-42 days) respectively. Eighteen individuals in this 130-player subgroup sustained a thumb metacarpal fracture. The return-to-play time was slower for patients sustaining thumb metacarpal fractures compared to other metacarpal fractures, and was significantly longer (median, 55 days) following nonsurgical treatment of thumb fractures compared with operative intervention (median, 24 days). A regression analysis revealed no trend or difference in return to football with respect to player age, time in the league, injury setting (practice vs game), injury mechanism, articular involvement, multiple concomitant injuries, or player position.

Conclusions: Most NFL players who sustain metacarpal fractures miss less than 3 weeks and return to play the same season. The only variables that lessen the return-to-play time are involvement of lesser digit metacarpals and operative intervention for treatment of thumb metacarpal fractures.

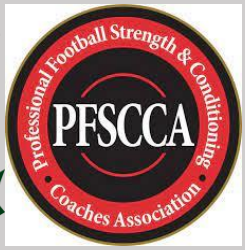
Type of study/level of evidence: Prognostic IV.

Monitoring external workload with wearable technology after anterior cruciate ligament reconstruction – A scoping review. The Orthopaedic Journal of Sports Medicine, 11(8): 1-10, 2023.

Background: Current sports medicine and rehabilitation trends indicate an increasing use of wearable technology. The ability of these devices to collect, transmit, and process physiological, biomechanical, bioenergy, and environmental data may aid in anterior cruciate ligament reconstruction (ACLR) workload monitoring and return-to-sport decision-making. In addition, their ease of use allows assessments to occur outside the clinical or laboratory settings and across a broader timeline.

Purpose: To (1) determine how wearable technology can assess external workload deficits between limbs (involved and uninvolved) and between groups (healthy controls vs patients with ACLR) during physical activity (PA) or sport and (2) describe the types of sensors, sensor specifications, assessment protocols, outcomes of interest, and participant characteristics from the included studies.

Study Design: Scoping review; Level of evidence, 4.



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Methods: In February 2023, a systematic search was performed in the MEDLINE, EMBASE, CINAHL, SPORTDiscus, Scopus, IEEE Xplore, Compendex, and ProQuest Dissertations and Theses Global databases. Eligible studies included assessments of PA or sports workloads via wearable technology after ACLR.

Results: Twenty articles met eligibility criteria and were included. The primary activity assessed was activities of daily living, although rehabilitation, training, and competition were also represented. Accelerometers, global positioning system units, pedometers, and pressure sensor insoles were worn to collect external workload data, which was quantified as kinetic, kinematic, and temporospatial data. Daily steps (count) and moderate to vigorous PA (min/day or week) were the most common units of measurement. A limited number of studies included outcomes related to between-limb asymmetries.

Conclusion: The findings of this scoping review highlight the versatility of wearable technologies to collect patients' kinetic, kinematic, and temporospatial data and assess external workload outcomes after ACLR. In addition, some wearable technologies identified deficits in workload compared with healthy controls and between reconstructed and unaffected limbs.

Performance-enhancing drugs in healthy athletes: An umbrella review of systematic reviews and meta-analyses. Sports Health, Ahead of Print: 1-11, 2023.

Context: Many clinicians, trainers, and athletes do not have a true understanding of the effects of commonly used performance-enhancing drugs (PEDs) on performance and health.

Objective: To provide an evidence-based review of 7 commonly used pharmacological interventions for performance enhancement in athletes.

Data Sources: PubMed and Scopus databases were searched on April 8, 2022.

Study Selection: Systematic reviews (SRs) and meta-analyses (MAs) assessing the performance-enhancing effects of the following interventions were included: androgenic anabolic steroids (AAS), growth hormone (GH), selective androgen receptor modulators (SARMs), creatine, angiotensin-converting enzyme (ACE)-inhibitors, recombinant human erythropoietin (rHuEPO), and cannabis.

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Study Design: Umbrella review of SRs and MAs.

Level of Evidence: Level 4.

Data Extraction: Primary outcomes collected were (1) body mass, (2) muscle strength, (3) performance, and (4) recovery. Adverse effects were also noted.

Results: A total of 27 papers evaluating 5 pharmacological interventions met inclusion criteria. No studies evaluating SARMs or ACE-inhibitors were included. AAS lead to a 5% to 52% increase in strength and a 0.62 standard mean difference in lean body mass with subsequent lipid derangements. GH alters body composition, without providing a strength or performance benefit, but potential risks include soft tissue edema, fatigue, arthralgias, and carpel tunnel syndrome. Creatine use during resistance training can safely increase total and lean body mass, strength, and performance in high-intensity, short-duration, repetitive tasks. Limited evidence supports rHuEPO benefit on performance despite increases in both VO2max and maximal power output, and severe cardiovascular risks are documented. Cannabis provides no performance benefit and may even impair athletic performance.

Conclusion: In young healthy persons and athletes, creatine can safely provide a performance-enhancing benefit when taken in controlled doses. AAS, GH, and rHuEPO are associated with severe adverse events and do not support a performance benefit, despite showing the ability to change bodily composition, strength, and/or physiologic measures. Cannabis may have an ergolytic, instead of ergogenic, effect.

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Plasma myoglobin indicates muscle damage associated with acceleration/deceleration during football. The Journal of Sports Medicine and Physical Fitness, Ahead of Print: 1-6, 2023.

Background: Monitoring muscle damage in athletes assists not only coaches to adjust the training workload but also medical staff to prevent injury. Measuring blood myoglobin concentration can help evaluate muscle damage. The novel portable device utilized in this study allows for easy on-site measurement of myoglobin, providing real-time data on the player's muscle damage. This study investigated the relationship between external load (global positioning system parameters) and internal loads (myoglobin concentration and creatine kinase activity) in 15 male professional football players before and after a match.

Methods: Whole blood samples from participants' fingertips were collected before the match (baseline) and at 2, 16, and 40 h after the match. Myoglobin concentrations were measured using the ia-100 compact immunoassay system. creatine kinase concentrations were measured in a clinical laboratory, and match loads were monitored using a global positioning system device.

Results: The mean myoglobin concentration was significantly higher at 2 h than at the other time points ($P<0.05$), and decreased to baseline levels within 16 h post-match. The mean creatine kinase concentration increased after the match but did not reach a significant level. Muscle damage monitored by myoglobin after football match-play was strongly associated with acceleration/deceleration metrics rather than the sprint/high-speed running distance.

Conclusions: Our findings indicate that myoglobin is a more sensitive marker of muscle damage than creatine kinase after football match play. Monitoring myoglobin in athletes can aid in determining their recovery status from the previous training load and help practitioners manage the training load.

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Postinjury alcohol use is associated with prolonged recovery after concussion in NCAA athletes. Clinical Journal of Sport Medicine, Ahead of Print: 1-8, 2023.

Objective: To determine whether alcohol use leads to prolonged clinical recovery or increased severity of concussion symptoms in National Collegiate Athletic Association (NCAA) athletes.

Design: Prospective observational study.

Setting: Clinical institutions.

Participants: Athletes from the NCAA Concussion Assessment Research and Education consortium who sustained a concussion from 2014 to 2021.

Interventions: Athletes were divided into 2 groups, those reporting alcohol use postinjury and those reporting no alcohol use postinjury.

Main Outcome Measures: Symptom recovery was evaluated as time (in days) from injury to clearance to return to unrestricted play (days until URTP). Severity of concussion symptoms was assessed using the Standardized Sport Concussion Assessment Tool (SCAT3) symptom severity, headache severity, difficulty concentrating, and difficulty remembering scores. These scores were taken a median of 6.6 [interquartile range (IQR) 5 4.0-10] and 6 (IQR 5 4.0-9.0) days after injury for those who did and did not consume alcohol postinjury respectively and compared with baseline SCAT3 scores.

Results: Four hundred eighty-four athletes from the data set had complete data for exposure and outcome. The adjusted mean number of days until URTP for athletes reporting alcohol use postinjury [23.3; 95% confidence interval (CI), 20.0-27.2; days] was incidence rate ratio (IRR) 1.32 (95% CI, 1.12-1.55; $P < 0.001$) times higher than for athletes who reported no alcohol use postinjury [17.7 (95% CI, 16.1-19.3) days]. Postinjury alcohol was not associated with severity of concussion symptoms (P's < 0.05).

Conclusion: Self-reported postinjury alcohol use is associated with prolonged recovery but not severity of concussion symptoms in collegiate athletes. This may inform future clinical recommendations regarding alcohol consumption after concussion.

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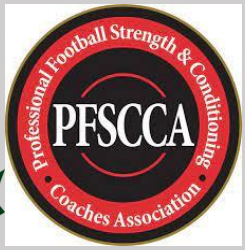

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Prophylactic knee bracing in offensive linemen of the National Football League – A retrospective analysis of usage trends, player performance, and major knee injury. The Orthopaedic Journal of Sports Medicine, 11(8): 1-8, 2023.

Background: Offensive linemen in American football are prone to high-energy valgus forces to the knee, leading to associated injuries. Some offensive linemen in the National Football League (NFL) wear prophylactic knee braces (PKB) to prevent ligamentous injury.

Purpose/Hypothesis: This purpose of the study was to compare injury rates and performance between NFL offensive linemen who wear PKB and those who do not. It was hypothesized that brace wear would be associated with fewer major knee injuries and no difference in gameplay performance.

Study Design: Cohort study; Level of evidence, 3.

Methods: For the 2014 through 2020 NFL seasons, offensive linemen with at least 200 game snaps per regular season were identified. Players were grouped by PKB status (bracers vs nonbracers) based on visualization of bilateral, dual-hinged metal knee braces as part of gameday uniforms on publicly available imaging databases and/or game videos. Major knee injuries, defined as those requiring the missing of games, were identified using publicly available data. Performance was assessed with Pro Football Focus grades for each season. Rates of major knee injury were compared between groups with the 2-sample Z test for proportions, and performance grades were compared with the unpaired t test.

Results: For the cumulative study period, bracers demonstrated a significantly lower rate of major knee injuries than nonbracers (0.013 vs 0.049 injuries per player, respectively; P = .04). Isolated MCL injury was the most common injury for nonbracers. There was no group difference in performance for the cumulative study period or during most individual seasons. Yearly prevalence of PKB usage declined steadily from 16.3% in 2014 to 5.6% in 2020. A subgroup analysis of rookie players demonstrated an overall downtrend in usage during the study period as well.

Conclusion: Results indicated that knee brace prophylaxis by NFL offensive linemen was associated with a reduced risk of major knee injury without a significant difference in performance when compared with nonbracers. Despite this, the prevalence of PKB declined over the study period.

Pro Football Sports Science Research Update

Proximal rectus femoris tendon tear in a professional football placekicker: A case report. Current Sports Medicine Reports, Vol. 22(6): 187-190, 2023.

Overview:

The rectus femoris (RF) is the most superficial quadriceps muscle. It acts as a hip flexor and knee extensor. Proximally, it has two tendon origins, the direct head attachment to the anterior inferior iliac spine (AIIS) and the indirect head attachment to the hip joint capsule at the superior acetabular ridge. The two heads come together to form a con joined tendon with the direct head fibers forming the superficial component and indirect head fibers forming the deep component.

The RF is the quadricep muscle most prone to injury given its biarticular nature. Injuries frequently result from a kicking or sprinting mechanism and are common in sports, such as football and soccer (3–6). Injuries often occur at the musculotendinous junction.

AIIS tendon avulsion injuries of the RF direct head are documented, but relatively uncommon. Proximal RF tears involving both the direct and indirect heads are even less frequently encountered. There are two prior cases in National Football League (NFL) punters who also performed placekicking duties with a similar mechanism and location of injury, but no cases of pure placekickers. This case describes the clinical features, management, and return to sport of a proximal RF tendon tear in a professional American style football placekicker.

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Pro Football Sports Science Research Update

Return to play following injuries in pro football (soccer): Insights into the real-life practices of 85 elite practitioners around diagnostics, progression strategies and reintegration processes. Sport Performance Science Reports, Vol. 1: 1-20, 2023.

Overview: Player injuries have a dramatic effect on pro football teams' performance, which carries important financial damages in parallel (i.e., an average EPL team loses approximately £ 45 million per season due to injury-related decrements in performance). Therefore, strategies aimed at reducing injury occurrence, and then accelerating to return to play and competition, is of primary interest to all practitioners working in elite teams. Following an injury, players enter into what is called the return-to-play (RTP) process, which is aimed at bringing them back to competition in the shortest time possible while reducing the risk of reinjury. The RTP process is generally composed of the following phases: diagnostic within hours to a few days following injury, treatment (healing phase) and injury-specific reconditioning (around the function of the injured structure), and finally generic and then football-specific conditioning that accompanies the return to training and competing. When it comes to enhancing these RTP protocols specifically, there have now been several key recommendations coming either from consensus statements (1) or Delphi surveys examining (hamstring) injury in professional football. These excellent pieces of research have improved our understanding of the likely optimal (scientific) process to follow, including specific screening protocols and suggested cut off values for progression criteria and overall RTP decision-making, i.e., when to return to run, play, and compete (e.g., eccentric hamstring strength relative to pre-injury benchmarks). However, a large proportion of the provided information is hamstring-specific, and while the majority of teams were reported to often meet the criteria that they set to progress through the RTP continuum, many practical, real-life questions and challenges that practitioners face daily have not been examined yet. It is well recommended that the RTP decision-making process should be shared between colleagues within the Multi Disciplinary Team (MDT) and with the technical staff and stakeholders. However, the actual processes within and outside the MDT in pro football have only been partially examined. Who makes the final decisions? How do practitioners from different departments (e.g., physio and conditioning coaches) share the workload and responsibilities with the injured players? How are diagnostics and delays communicated and followed up with coaches? How often do players request an opinion from an external-to-the-club specialist? How often do players request to be trained by an external practitioner? How flexible are practitioners when it comes to actually following the ideal/theoretical progression

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protocols? Does player status and pressure from coaches and calendars affect all of the above? Simple but highly recurrent logistical questions also deserve more examination, e.g., when do injured players train in relation to the main squad? Do they need to participate in team meetings? In the daily life of a MDT in a pro football club, all the above questions are often bigger pain points that strictly follow progression guidelines.

Aim of the Study: This survey has been designed to gain insight into how practitioners' approach, manage, and deliver the "return to play" process (RTP, as the overall rehabilitation process that leads to the generic idea of returning to 'play'), and how MDT approach the process and decision-making in the real-life conditions of elite football (soccer). More precisely, our aim was to answer key practical questions around clinical diagnosis, the use of progression criteria, players' reintegration with the group/competition, and finally, the overall organization of the RTP process within the MDT.

Return-to-play for elite athletes with genetic heart diseases predisposing to sudden cardiac death. Journal of the American College of Cardiology, Vol. 82(8): 661-670, 2023.

Background: People diagnosed with genetic heart diseases (GHDs) associated with sudden cardiac death (SCD) have historically been restricted from competitive sports. Recent data documenting return-to-play (RTP) experiences following shared decision making (SDM) suggest that cardiac event rates for athletes with a GHD are lower than previously described, thereby suggesting an opportunity to reconsider this paradigm.

Objectives: The purpose of this study was to evaluate clinical outcomes among National Collegiate Athletic Association Division I university and professional athletes diagnosed with a GHD.

Methods: A multicenter retrospective analysis was performed to examine demographics, clinical characteristics, RTP outcomes, and cardiac events among elite athletes with a GHD. RESULTS A total of 76 elite (66%, Division I, 34% professional) athletes (age 19.9 ± 5 years, 28% women) diagnosed with a GHD (hypertrophic cardiomyopathy [53%], long QT syndrome, long QT syndrome [26%]) comprise this cohort. Most athletes were asymptomatic (48 of 76, 63%) before diagnosis and had their GHD detected during routine preparticipation cardiovascular screening. Most athletes (55 of 76, 72%) were initially disqualified from their sport but subsequently opted for unrestricted RTP after comprehensive clinical

evaluation and SDM. To date, (mean follow-up 7 ± 6 years), only 1 exercise-related (1.3%) and 2 nonexercise-related GHD-associated adverse cardiac events occurred. There have been no fatalities during follow-up.

Conclusions: This is the first study describing the experience of athletes with a known SCD-predisposing GHD who are competing at the elite level. After careful evaluation, risk stratification, and tailoring of their GHD therapy, RTP following SDM appears associated with low, nonfatal events rates at elite levels of sport.

SARS-CoV-2 infection increases the risk of muscle injury in professional male soccer players – A retrospective analysis of the Italian and Spanish major leagues. Infectious Disease Reports, 15: 425-435, 2023.

Abstract: A retrospective cohort study on professional soccer players from the Serie A and LaLiga was conducted to investigate the correlation between SARS-CoV-2 infection and muscle injuries. Players were divided into two groups based on whether they contracted the SARS-CoV-2 infection (C+) or not (C−) during the 2020/2021 season. In the 2019–2020 season, both championships showed a similar number of muscular injuries (MI) between C+ and C− (Serie A: $p = 0.194$; 95% CI: −0.044 to 0.215, LaLiga $p = 0.915$; 95% CI: −0.123 to 0.137). In the 2020–2021 season, C+ had a significantly higher number of MI compared to C− in both championships (Serie A: $p < 0.05$; 95% CI 0.731 to 1.038; LaLiga: $p < 0.05$; 95% CI: 0.773 to 1.054). Multiple linear regression analysis confirmed that belonging to C+ in the season 2020/2021 was the variable that most strongly influenced the probability of having a muscle injury. Survival analysis revealed a hazard ratio of 3.73 (95% CI 3.018 to 4.628) and of 5.14 (95% CI 3.200 to 8.254) for Serie A and LaLiga respectively. We found an association between SARS-CoV-2 infection and increased risk of muscle injury, emphasizing the importance of carefully considering the infection in the decision-making process for returning to sport. Therefore, SARS-CoV-2 infection should be judged as a real injury requiring specific assessment and training programs.

Pro Football Sports Science Research Update

Vitamin C supplementation and athletic performance: A review. Current Sports Medicine Reports, Vol. 22(7): 255-259, 2023.

Abstract: Many athletes utilize high-dose vitamin C supplementation to optimize athletic performance. A review of research over the past 10 years on the use of vitamin C and athletic performance show mixed results. Fourteen randomized control trials were reviewed. In most studies, vitamin C was used with at least one additional supplement, usually vitamin E. Three studies showed positive outcomes associated with decreased markers of muscle damage after intense exercise with some form of vitamin C supplementation. The remaining 11 articles showed either neutral or negative effects of high dose vitamin C supplementation on muscle damage, physical performance, perceived muscle soreness, and/or adaptations to training. Based on a lack of consistent data and potential for blunted physiologic adaptations to training, long-term high-dosage supplementation with vitamin C is not recommended. Athletes should obtain antioxidants through a nutrient-rich diet instead of through supplement use.

The impact of daytime napping following normal night-time sleep on physical performance: A systematic review, meta-analysis and meta-regression. Sports Medicine, Ahead of Print: 1-23, 2023.

Background: Daytime napping is used by athletes as a strategy to supplement night time sleep and aid physical performance. However, no meta-analytical overview regarding the impact of napping following a night of normal sleep (7–9 h) on physical performance is available.

Objective: The aim of this study was to evaluate the effect of daytime napping following normal night-time sleep on physical performance in physically active individuals and athletes.

Methods: This systematic review and meta-analysis was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines. Seven electronic databases (i.e., PubMed, Web of Science, Scopus, SPORTDiscus, CINAHL, SCIELO, and EBSCOhost) were used to search for relevant studies that investigated the impact of daytime napping, following normal night-time sleep, on physical performance in physically active individuals and athletes, published in any language, and available before September 01, 2022. Studies that included assessments of any physical

performance measures were included. QualSyst was used to assess the methodological quality of the studies.

Results: Of 18 selected articles, 15 were of strong quality and 3 were of moderate quality. Compared with no-nap conditions, physically active individuals and athletes who napped experienced an increase in highest distance (effect size [ES] 1.026; $p<0.001$) and total distance (ES 0.737; $p<0.001$), and a decrease in fatigue index (ES 0.839, $p=0.008$) during the 5-m shuttle run test (5MSRT). However, napping yielded no effect on muscle force (ES 0.175; $p=0.267$). No effect of napping was found in one study that measured sprint performance and in two studies that measured performance during the 30-s Wingate test. Two of three studies reported an increase in jump performance after napping. Two of three studies reported an increase in repeated sprints after napping. One study reported an increase in upper-body power performance after napping, and napping was beneficial for endurance performance in one of two studies.

Conclusion: Following normal sleep, napping is beneficial for the performance of the 5MSRT, with no significant effect on muscle force. No firm conclusions can be drawn regarding other physical performance measures due to the limited number of studies.

Key Points

- Daytime napping before afternoon training sessions and/or competition could be recommended as a way to supplement night-time sleep as well as enhance athletic performance.
- Longer naps might be more beneficial in optimizing physical performance.
- The impact of a diurnal nap may be affected by (i) nap durations, (ii) time of day of naps, (iii) sleep inertia, and (iv) exercise type.